

# Platform saturation, tourist amenities, and Airbnb prices in European cities

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## Introduction

It is well known that Airbnb prices vary systematically across listings (uploaded housing offers), neighborhoods, and cities. A certain part of this variation may obviously be attributed to the features of the listings themselves, such as accommodation type, size, quality, and host characteristics. Still, some part of the remaining variation may be explained with the differences between neighborhoods themselves: two otherwise very similar listings may be priced differently depending on where they are located.

In the present analysis, I explore how Airbnb prices are affected by listing characteristics, neighborhood characteristics, and city-level differences in seven European cities: Amsterdam, Berlin, London, Madrid, Rome, Venice, and Vienna. The **dependent variable** is the logarithm of nightly price. As for the **explanatory variables**, I focus on the three factors: listing characteristics (for example, whether a listing is an entire home or a private room, etc.), neighborhood Airbnb density, and tourist-amenity density (the density of tourist amenities such as restaurants, cafes, bars, museums, attractions, theaters, and other places like that within a neighborhood). I also include controls like host characteristics, neighborhood share of entire-home listings, distance to the city center, and city-level random effects.

The **research question** therefore is:

How are Airbnb prices shaped by listing characteristics, neighborhood tourist amenities, neighborhood Airbnb saturation, and city-level differences across European urban tourism markets?

To answer this question, I use **three datasets**. First, I employ detailed listing data from Inside Airbnb to measure prices and listing characteristics. Then, I also use Inside Airbnb neighborhood boundary data to assign each listing to a neighborhood and to calculate neighborhood area and distance to the city center. Finally, I use points of interest from the OpenStreetMap cartographic data to measure tourist-amenity density. As already mentioned, tourist amenities refer to the mapped places that are likely to be useful or attractive for short-stay visitors: restaurants, cafes, bars, pubs, museums, attractions, galleries, theaters, cinemas, viewpoints, monuments, and so on. For each neighborhood, I count these amenities and divide by the neighborhood area.

The **main goal** of my analysis is to distinguish between two closely related but conceptually different neighborhood characteristics. The first is Airbnb density, measured as the number of Airbnb listings per square km, which represents the local concentration of Airbnb supply. The second is tourist-amenity density, measured as the number of tourist-adjacent amenities per square km, which represents the local concentration of tourist-adjacent urban infrastructure. Airbnb density therefore describes the scale of the local short-term rental market, whereas tourist-amenity describes the surrounding environment that visitors consume during their stay.

**Multilevel linear models** are used to empirically address the goal. Listings are nested within neighborhoods, and neighborhoods are nested within cities: listings in the same neighborhood share local conditions, and neighborhoods in the same city share broader market conditions. In the models I first estimate how much price varies across neighborhoods and cities, then I add listing-level controls, then neighborhood-level contextual variables, and finally I test whether the price difference between entire-home listings and private room listings varies across neighborhoods.

The analysis therefore shows how much of Airbnb price variation is linked to the listing itself and how much is linked to neighborhood and city context. An addition of the independent tourist-amenities measure makes it possible to estimate whether high Airbnb prices are associated with platform saturation itself, with tourist-adjacent places, or with both.

## **Theoretical background and Hypotheses**

### **Accommodation characteristics and hedonic pricing**

In the hedonic price theory, Rosen (1974) argues that the price of a product reflects the value of its individual characteristics. In housing and accommodation markets, people pay for specific features such as size, quality, location, and comfort. Airbnb listings therefore are a clear example of this type of market because guests choose between listings with different characteristics and pay different prices for them: entire homes offer more privacy than private rooms, bigger listings can accommodate more guests, and highly rated (or superhost) listings may increase the trust level. These characteristics should therefore be associated with higher prices.

Just mentioned expectation is also consistent with research on Airbnb as an accommodation market. Guttentag (2015) argues that Airbnb does not simply provide a cheaper alternative to hotel accommodation; it offers a different type of accommodation, often based on home-like space and variety. Zervas et al. (2017) show that Airbnb reduces hotel revenue in some parts of the accommodation market more than in others: the largest declines are among low-priced hotels and during periods of high demand, suggests that guests do not treat all accommodation options as interchangeable and that differences in accommodation type may be important for pricing.

This leads to the first two hypotheses:

- H1.** Entire-home listings will have higher prices than private-room listings, holding other listing and neighborhood characteristics constant.
- H2.** Listings with better accommodation and reputation characteristics (capacity, bedrooms and bathrooms, higher ratings, host trust signals) will have higher prices.

### **Location and tourist amenities**

For short-term accommodation, location is part of the product being sold. Short-term visitors usually spend only a few days in a city and might choose accommodation partly to gain easier access to attractions, restaurants, cafes, nightlife, museums, etc. A neighborhood that offers many such opportunities is therefore likely to produce higher accommodation prices. This expectation follows from urban economics research, which shows that people are willing to pay for access to local amenities and dense urban environments (Glaeser et al., 2001). In general, economic value tends to concentrate where amenities and activities are clustered, leading to higher prices in those locations (Rosenthal & Strange, 2004).

Several studies support just mentioned ideas in the context of Airbnb specifically. Gutierrez et al. (2017) compare Airbnb and hotel locations in Barcelona and show that Airbnb listings are disproportionately concentrated near major tourist attractions and in neighborhoods with higher tourist activity. Stors and Baltes (2018) show that Airbnb hosts often mention/market nearby places in their listings, meaning that neighborhood amenities play a role how the accommodation is presented to potential guests. De Nadai and Lepri (2018) also show that features of the surrounding urban environment, including walkability and the density of local amenities and services, help explain real estate value.

If tourist amenities contribute to the value that guests derive from a location, Airbnb hosts should therefore be able to establish bigger prices in neighborhoods where such amenities are more common.

This leads to the third hypothesis:

**H3.** Listings in neighborhoods with higher tourist-amenity density will have higher nightly prices, holding listing characteristics, Airbnb density, and city differences constant.

Distance from the city center represents another aspect of location that is slightly different from the tourist-amenity density. Even neighborhoods with similar amenity levels may differ in how central they are within the city. Since central locations are generally more attractive to visitors and provide easier access to urban destinations, listings farther from the city center are therefore expected to have lower prices, all else equal.

**H4.** Listings located farther from the city center will have lower prices.

## **Airbnb density and neighborhood commercialization**

Moreover, neighborhoods differ not only in the number of tourist amenities they have but also in the size and structure of their Airbnb market. For example, two neighborhoods may have a similar number of restaurants, cafes, attractions, etc., but one may contain only a small number of Airbnb listings while the other may have a lot of them. Airbnb density captures this difference, as it measures how concentrated Airbnb activity is in a neighborhood.

Previous research shows that Airbnb listings tend to cluster in particular parts of cities rather than being evenly distributed across urban space (Quattrone et al., 2016). However, it is also important to consider what kinds of listings make up the local Airbnb market. Wachsmuth and Weisler (2018) argue that many Airbnb markets get increasingly more dominated by entire-home rentals, which are realistically closer to commercial tourist accommodation than to traditional home sharing.

To explore these two aspects of the local Airbnb market, I introduce two variables in the analysis. Airbnb density measures the number of Airbnb listings per square km, and captures how large and concentrated the local Airbnb market is. Neighborhood share of entire-home listings measures the proportion of Airbnb listings in the neighborhood that are entire homes rather than private rooms, and captures whether that market is mainly based on entire-home rentals or on room sharing. Including both of them allows to distinguish between the scale of Airbnb activity and the type of Airbnb market present in a neighborhood.

This leads to the fifth hypothesis:

**H5.** The neighborhood share of entire-home listings will be positively associated with price, whereas the independent effect of Airbnb density will weaken once tourist-amenity density is included.

## **Why the price difference between entire homes and private rooms may vary across neighborhoods**

Finally, entire homes and private rooms do not necessarily exist under the same market conditions everywhere: differences in local demand, visitor profiles, and the composition of the Airbnb market may affect how much guests are willing to pay for exclusive use of a property.

To account for this possibility, in the analysis I allow the effect of entire-home status to vary across neighborhoods, and test whether this variation is related to Airbnb density. Previous research shows that Airbnb markets are spatially uneven and that competitive conditions differ across locations and market segments (Quattrone et al., 2016; Zervas et al., 2017). This suggests that the price advantage of entire-home listings may depend on neighborhood context rather than being constant across the city.

This leads to the final hypothesis:

**H6.** The price premium for entire-home listings will vary across neighbourhoods and will be larger in neighbourhoods with higher Airbnb density.

## **Empirical strategy**

The hypotheses inform how the modeling sequence is built. The empty model first estimates how much price variation there is between listings, neighborhoods, and cities. The listing-level model tests H1 and H2 by estimating whether accommodation type, size, ratings, and host characteristics are associated with price. The contextual model then adds tourist-amenity density, distance to the city center, Airbnb density, and neighborhood share of entire homes to test H3, H4, and H5. Finally, the random-slope and interaction models are fit to test H6 by allowing the entire-home price benefit to differ across neighborhoods and by estimating whether it changes as Airbnb density increases.

## **Data and variables**

### **Data sources and scope**

In the analysis, I employ three sources of data: Airbnb listing data, neighborhood boundary data, and OpenStreetMap points of interest data.

Inside Airbnb detailed listings files (Inside Airbnb, 2025) are used to retrieve nightly prices, room types, accommodation capacity, bedrooms, bathrooms, reviews, ratings, host characteristics, availability measures, and listing coordinates. Nightly price is the dependent variable (`log_price`), whereas room type (`entire_home`), accommodation capacity (`accommodates`), bedrooms, bathrooms, review measures, host characteristics (e.g., `host_is_superhost`), and booking features (e.g., `instant_bookable`) are included as listing-level predictors in the models.

Moreover, Inside Airbnb neighborhood GeoJSON files (Inside Airbnb, 2025) are used to assign listings to neighborhoods and to calculate neighborhood-level characteristics such as area and distance to the city center.

To count the local visitor-oriented facilities, I use the OpenStreetMap data on points of interest (OpenStreetMap Contributors, 2026; Pebesma & Bivand, 2023; Padgham et al., 2017), where I collect

information on restaurants, cafes, bars, museums, attractions, theaters, monuments, and similar amenities and aggregate the counts to the neighborhood level to create the tourist-amenity density variable.

Initially, I retrieved data for the ten European cities: Amsterdam, Barcelona, Berlin, Lisbon, London, Madrid, Paris, Rome, Venice, and Vienna. However, Barcelona, Lisbon, and Paris had to be excluded because their listing and calendar files did not contain nightly price values (the entire columns were N/A). The final sample therefore includes only seven cities: Amsterdam, Berlin, London, Madrid, Rome, Venice, and Vienna.

All seven cities are big European tourism destinations, which differ in their average price levels, urban size, spatial patterns of tourism, and the composition of their Airbnb markets. Those differences produce variation at both the city and neighborhood levels, making it possible to explore with multilevel modeling how Airbnb prices vary across listings, neighborhoods, and cities, and whether neighborhood characteristics explain price differences after accounting for listing-level variables.

## Level of analysis and the outcome variable

The unit of analysis is the Airbnb listing. Each listing belongs to one neighborhood and one city. The dependent variable is the **natural logarithm of nightly price** in euros. I used the log transformation to fix the right-skewness of the data.

London's prices were converted from pounds to euros with a fixed conversion factor. To reduce the influence of extreme observations, I trimmed prices for each city at the 1st and 99th percentiles. Also, I only kept two property types in the analysis: entire homes/apartments and private rooms. Shared rooms and hotel rooms represent different accommodation categories and form a small part of the data, so including them would introduce additional sources of price variation unrelated to the comparison between entire homes and private rooms, hence my decision to exclude them.

## Listing-level variables

The model includes the following listing-level predictors:

- `entire_home`: a binary indicator equal to 1 for entire home/apartment listings and 0 for private rooms;
- number of guests accommodated: standardized number of guests the listing accommodates;
- number of bedrooms: standardized number of bedrooms;
- number of bathrooms: standardized number of bathrooms;
- review score: standardized review score rating;
- number of reviews: standardized log number of reviews;
- host superhost status: 1 if the host is a superhost, 0 otherwise;
- instant bookability (whether guests can book the listing immediately without host approval): 1 if the listing can be booked instantly, 0 otherwise;
- minimum nights requirement: standardized minimum stay requirement;

- availability over 365 days: standardized number of days the listing is available during the next 365 days.

Listing type affects the degree of privacy and space offered to guests, therefore I treat it as one of the most important predictors of the Airbnb prices. The remaining variables record differences in listing size, quality, host characteristics, and booking conditions that may also influence nightly prices. Before modeling, I standardize continuous predictors.

## Neighborhood-level Airbnb variables

I use two Airbnb neighborhood variables.

The first is **Airbnb density**, which is the number of Airbnb listings in a neighborhood divided by the neighborhood area in square km. As Airbnb density is used to explain the price of individual listings, I exclude the listing itself when calculating the density value assigned to that listing. For each listing, I calculate neighborhood Airbnb density using all other listings in the same neighborhood and exclude the current listing itself (leave-one-out approach).

The second variable is the **neighborhood share of entire-home listings**. This measures the composition of the local Airbnb market. A high value means that a given neighborhood's Airbnb market consists mostly of entire homes rather than private rooms. I also calculate this variable using a leave-one-out approach.

## Neighbourhood-level tourist-amenity density

**Tourist-amenity density** measures the number of visitor-oriented amenities per square km in each neighborhood. I use OpenStreetMap points of interest and count amenities that are likely to be interesting for the short-stay visitors: restaurants, cafes, bars, pubs, museums, attractions, galleries, theaters, cinemas, viewpoints, monuments, memorials, castles, archaeological sites, artwork installations, fountains, zoos, theme parks, aquariums, beaches, marinas, marketplaces, and information points. I log-transform and standardize this variable before adding it in the models.

## Distance to the city center

I calculate **distance to the city center** using neighborhood geometry. For each neighborhood, I calculate the distance between a representative point in the neighborhood and a manually selected city center coordinate (importantly, those are arbitrary, my choice was based on the historical/functional significance but ultimately it depends on one's "city center" definition). I use "Alexanderplatz" for Berlin, "Piazza Venezia" for Rome, "Piazza San Marco" for Venice, "Puerta del Sol" for Madrid, "Dam Square" for Amsterdam, "Charing Cross" for London, and "Stephansplatz" for Vienna as reference points. In general, this variable is included because accessibility to central locations is commonly reflected in prices (Alonso, 2013). I standardize distance within each city because they differ in size. A one-unit increase in `dist_center_z` means that a neighborhood is farther from the center relative to other neighborhoods in the same city.

## Models

The data has hierarchical structure: listings are located within neighborhoods, and neighborhoods are located within cities. I therefore estimate multilevel linear models with random intercepts for both neighborhoods and cities, so that the specification accounts for the fact that listings in the same neighborhood share local conditions, while neighborhoods in the same city share broader market conditions.

The empty model (**model 0**) estimates price variation before adding explanatory variables:  $\beta_0 + u_{0k} + v_{0jk} + e_{ijk}$ , where  $u_{0k}$  is the city-level random intercept,  $v_{0jk}$  is the neighborhood-level random intercept, and  $e_{ijk}$  is the listing-level residual.

I then estimate a sequence of models corresponding to the hypotheses.

In **model 1**, listing-level predictors are added, such as room type, accommodation capacity, bedrooms, bathrooms, ratings, reviews, host characteristics, minimum nights, and availability. This model tests whether listing characteristics explain price differences.

In **model 2**, neighborhood-level predictors are added, such as Airbnb density, neighborhood share of entire-home listings, tourist-amenity density, mean neighborhood reviews, mean neighborhood availability, and distance to the city center. This model tests whether neighborhood context is associated with price after controlling for listing characteristics.

In **model 3**, I allow the effect of being an entire-home listing to be different across neighborhoods. In the previous models, the price difference between an entire home and a private room is assumed to be the same everywhere, so in Model 3 this assumption is freed by allowing the price advantage of entire-home listings over private rooms to differ across neighborhoods. In some neighborhoods, guests may be willing to pay a much larger prices for privacy and exclusive use of the property, while in other neighborhoods the difference between an entire home and a private room may be smaller.

**Model 4** is used to test whether the relationship between Airbnb density and price differs between entire-home listings and private rooms. To do this, I add an interaction between listing type and neighborhood Airbnb density. This model examines whether the entire-home price benefit gets larger or smaller in neighborhoods with higher concentrations of Airbnb listings.

**Model 5** is the final model, which combines the interaction from Model 4 with the neighborhood-level random slope from Model 3. This model is the most complete specification as it allows the entire-home premium to vary across neighborhoods and also tests whether part of that variation is related to neighborhood Airbnb density.

I estimate the models using maximum likelihood so that model fit can be compared across specifications.

## Exploratory diagnostics

The final dataset has 143,159 listings. Figure 1 shows median Airbnb price by city. Amsterdam has the highest median price, followed by Venice and London. Berlin, Vienna, and Madrid have lower median prices. These differences support the use of city-level random intercepts.

Figure 2 shows the relationship between neighborhood Airbnb density and median nightly price. The relationship seems to be positive in several cities, such as Amsterdam, London, Madrid, Rome, and Venice. This pattern fits the expectation that Airbnb listings cluster in central and tourist-oriented urban areas (the figure is descriptive).

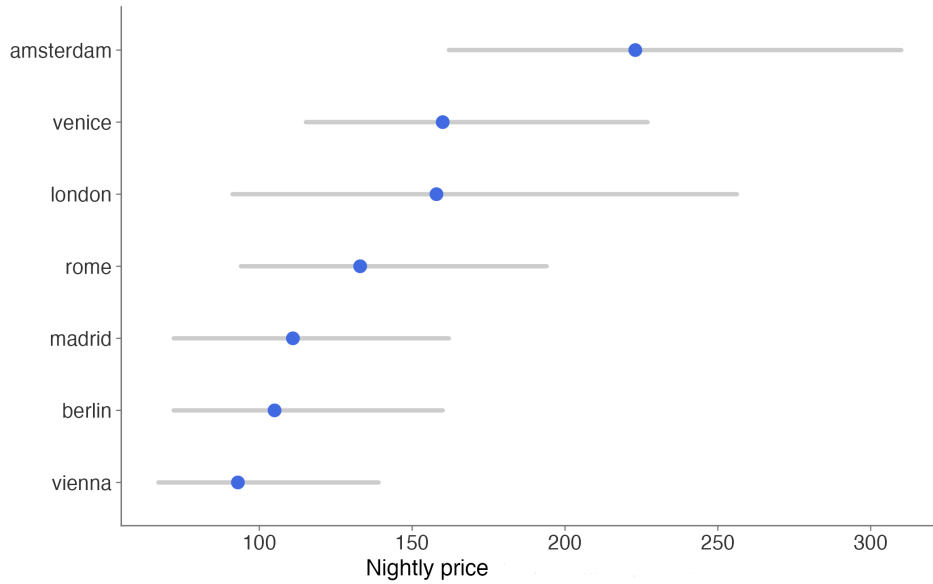


Figure 1: Median Airbnb nightly price by city

Figure 3 shows the relationship between tourist-amenity density and median nightly price. Neighborhoods with more tourist amenities per square km tend to have higher median Airbnb prices. The relationship is visible in most cities. It seems to suggest that the surrounding tourist-adjacent environment affects the price (the figure is descriptive).

## Results

Table 1 contains the main coefficients from the final model (M5) needed to answer the hypotheses. Table 2 has contains coefficients for all models and is included in the appendix.

### Listing-level predictors

The listing-level results support H1 and H2: entire-home listings are more expensive than private rooms in every model. In the final model, the coefficient is 0.459. This corresponds to approximately:  $100 \times (e^{0.459} - 1) \approx 58\%$ . Entire-home listings therefore have about 58% higher nightly prices than private rooms, holding the other variables constant and evaluating the effect at average Airbnb density.

Other listing-level variables also behave as expected: listings that accommodate more guests have higher prices. Listings with more bedrooms and bathrooms also have higher prices. Higher ratings, superhost status, instant bookability, and higher availability are positively associated with price. The coefficient for log reviews is negative, which may mean that highly reviewed listings are older, more frequently booked, or priced more competitively.

### Neighborhood tourist amenities

Moreover, the results strongly support H3. Tourist-amenity density is positive and statistically significant in every contextual model. In the final model, the coefficient is 0.129, which means that one standard deviation increase in logged tourist-amenity density is associated with approximately  $100 \times (e^{0.129} - 1) \approx 13.8\%$

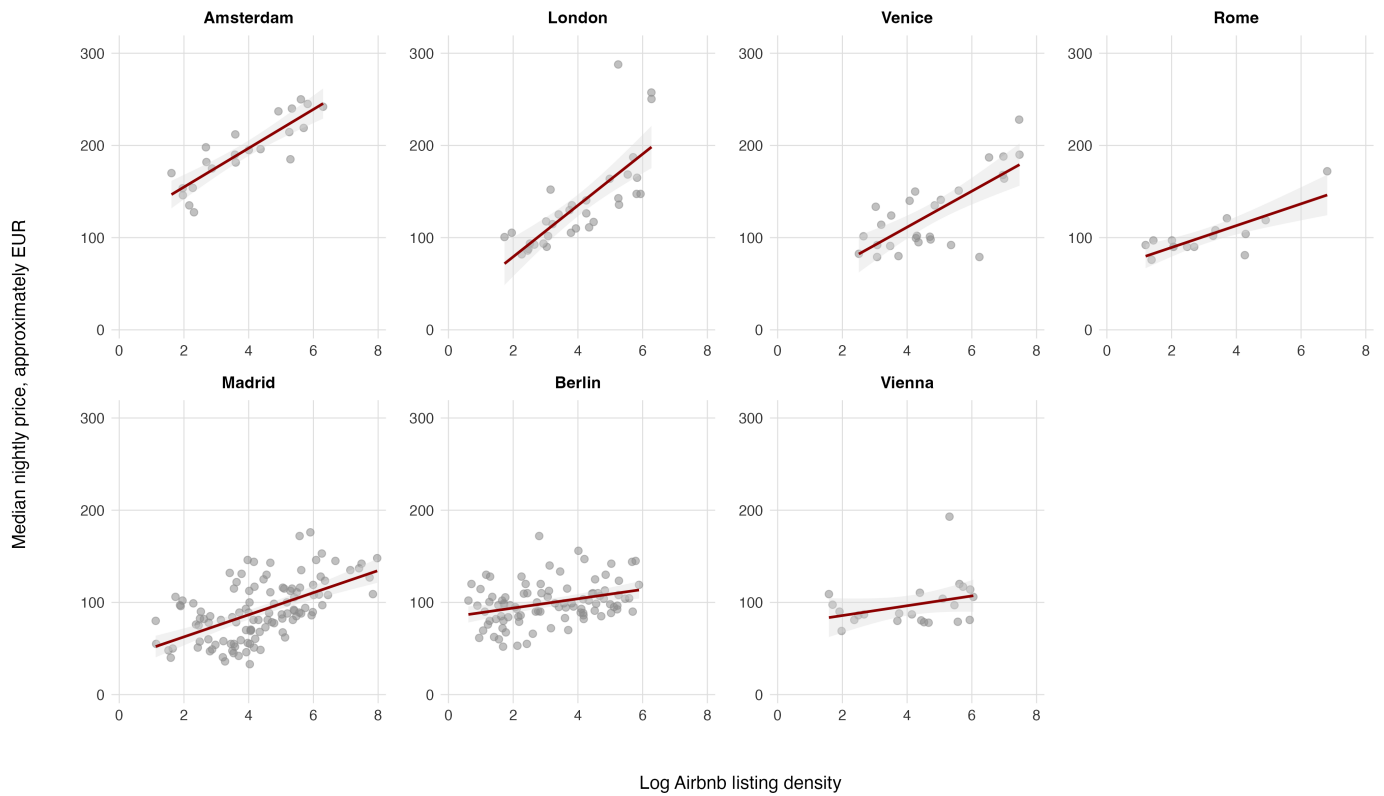


Figure 2: Neighbourhood Airbnb density and median nightly price

higher nightly price, holding listing characteristics, Airbnb density, neighborhood share of entire homes, distance to the center, and city and neighborhood clustering constant.

Airbnb listings therefore cost more in neighborhoods with more visitor-oriented amenities, which supports the assumption that the surrounding urban environment partly defines what visitors pay for.

## Distance to the center

H4 is also supported: the distance to the city center is negative and statistically significant. In the final model, the coefficient is  $-0.042$ . Since distance is standardized within city, this means that listings in neighborhoods farther from the center of their own city tend to have lower prices.

## Airbnb density and neighborhood composition

H5 is partially supported. The neighborhood share of entire-home listings is positive and statistically significant: in the final model the coefficient is  $0.036$ , which means that listings in neighborhoods where the Airbnb market contains more entire homes tend to have higher prices, even after controlling for whether the individual listing is itself an entire home.

Airbnb density has a different pattern, however: in the descriptive plot (Figure 2), density and price seem to have a positive relationship in several cities. In the multilevel model, after tourist-amenity density is included, the coefficient for Airbnb density becomes small and negative. In the final model, the coefficient is  $-0.046$  and modestly statistically significant.

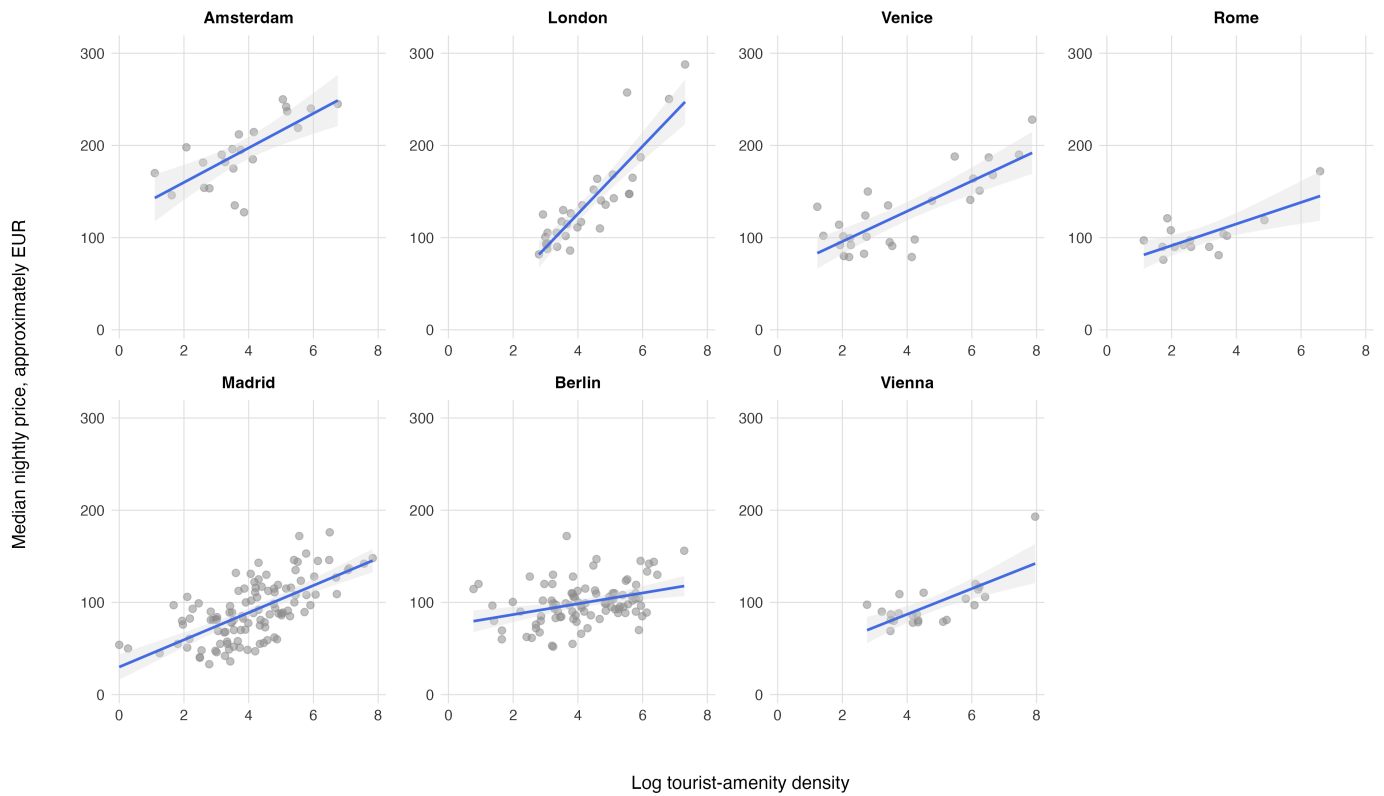


Figure 3: Tourist-amenity density and median Airbnb nightly price

I suppose that positive descriptive association between Airbnb density and price visible in the Figure 2 partly reflects the concentration of tourist amenities. Neighborhoods with many Airbnb listings often also contain many visitor-oriented amenities, and once those amenities are included in the model, Airbnb density no longer captures only tourist attractiveness, it may also reflect local competition between listings. For private rooms, higher Airbnb density is associated with slightly lower prices, a pattern that is consistent with the interaction result discussed below.

### Variation in the price advantage of entire-home listings

The results support H6. The random-slope models show that the entire-home price advantage varies across neighborhoods. In M3 and M5, the model allows the effect of entire-home status to differ by neighborhood, and model fit improves substantially compared with the random intercept only contextual model. The conditional ( $R^2$ ) increases from 0.612 in M2 to 0.645 in M3 and 0.653 in M5, which shows that the neighborhood-level variation in the price advantage of entire-home listings over private rooms does exist.

The interaction between entire-home status and Airbnb density is positive in the final model with the coefficient of 0.033. Since the main effect of Airbnb density refers to private rooms, it means that the density-price relationship is less negative for entire homes than for private rooms. In other words, entire homes have a better price advantage in Airbnb-dense neighborhoods.

M3 has a slightly lower AIC than M5, so the interaction does not necessarily improve model fit. The interaction is nevertheless statistically significant and shows that the price difference between entire homes and private rooms depends partly on neighborhood Airbnb density. At the same time, the random-slope result shows substantial variation in the entire-home price advantage across neighborhoods.

Table 1: Final model results (for all models comparison please refer to the Table 2 in the appendix)

| Predictor                           | Coefficient | Standard error |
|-------------------------------------|-------------|----------------|
| Entire home                         | 0.459***    | 0.016          |
| Accommodates                        | 0.153***    | 0.002          |
| Bedrooms                            | 0.094***    | 0.002          |
| Bathrooms                           | 0.062***    | 0.001          |
| Rating                              | 0.044***    | 0.001          |
| Review missing                      | 0.013**     | 0.004          |
| Log reviews                         | -0.085***   | 0.002          |
| Superhost                           | 0.076***    | 0.003          |
| Instantly bookable                  | 0.067***    | 0.002          |
| Log minimum nights                  | -0.049***   | 0.001          |
| Availability 365                    | 0.037***    | 0.001          |
| Airbnb density                      | -0.046*     | 0.022          |
| Neighborhood share entire home      | 0.036***    | 0.009          |
| Mean neighborhood reviews           | 0.032*      | 0.014          |
| Mean neighborhood availability      | 0.025       | 0.013          |
| Tourist-amenity density             | 0.129***    | 0.020          |
| Distance to the center              | -0.042**    | 0.016          |
| Entire home $\times$ Airbnb density | 0.033**     | 0.012          |
| Observations                        |             | 143,159        |
| Marginal $R^2$                      |             | 0.426          |
| Conditional $R^2$                   |             | 0.653          |
| AIC                                 |             | 155587.0       |
| RMSE                                |             | 0.41           |

Note: the dependent variable is log-transformed nightly price. The model includes random intercepts for city and neighborhood and a neighborhood-level random slope for entire-home status. \*  $p < 0.05$ , \*\*  $p < 0.01$ , \*\*\*  $p < 0.001$ .

## Conclusion

In my analysis, I explored the Airbnb prices across seven European cities using a three-level multilevel model with listings nested within neighborhoods and neighborhoods nested within cities. The results show that both listing characteristics and neighborhood context contribute to the price variation.

At the listing level, entire homes tend to have substantially higher prices than private rooms. Bigger listings and better quality characteristics are also associated with higher prices. At the neighborhood level, tourist-amenity density is the strongest contextual predictor: listings located in areas with higher concentrations of restaurants, cafes, bars, museums, attractions, theaters, and similar amenities are priced significantly higher, even after controlling for listing characteristics and city-level differences.

The results also show in what ways Airbnb density and tourist attractiveness are different. Once tourist-amenity density is included in the model, Airbnb density no longer functions as a positive indicator of neighborhood value. This suggests that the concentration of Airbnb listings and the concentration of visitor-oriented amenities represent different dimensions of the urban space. Neighborhood prices are shaped not only by the presence of Airbnb activity but also by the broader environment available to the visitors.

The random slope models further show that the price advantage associated with entire-home listings is different across neighborhoods. The value of listing type is therefore not constant across urban contexts but depends partly on local market conditions.

Overall, the findings suggest that Airbnb prices are influenced not only by the characteristics of individual properties but also by the contexts in which those properties are located: they are embedded in neighborhood environments, shaped by the distribution of tourist amenities, local Airbnb market composition, and broader urban geography.

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## Appendix (full models table below)

Table 2: Full multilevel model results

|                                   | M0 Empty            | M1 Listing controls  | M2 Context + OSM     | M3 Random slope + OSM | M4 Interaction + OSM | M5 Final + OSM       |
|-----------------------------------|---------------------|----------------------|----------------------|-----------------------|----------------------|----------------------|
| Intercept                         | 4.759***<br>(0.096) | 4.401***<br>(0.097)  | 4.500***<br>(0.103)  | 4.559***<br>(0.109)   | 4.500***<br>(0.103)  | 4.556***<br>(0.108)  |
| Entire home                       |                     | 0.433***<br>(0.003)  | 0.433***<br>(0.003)  | 0.436***<br>(0.013)   | 0.433***<br>(0.003)  | 0.459***<br>(0.016)  |
| Accommodates                      |                     | 0.160***<br>(0.002)  | 0.160***<br>(0.002)  | 0.153***<br>(0.002)   | 0.160***<br>(0.002)  | 0.153***<br>(0.002)  |
| Bedrooms                          |                     | 0.099***<br>(0.002)  | 0.099***<br>(0.002)  | 0.094***<br>(0.002)   | 0.099***<br>(0.002)  | 0.094***<br>(0.002)  |
| Bathrooms                         |                     | 0.052***<br>(0.001)  | 0.052***<br>(0.001)  | 0.062***<br>(0.001)   | 0.052***<br>(0.001)  | 0.062***<br>(0.001)  |
| Rating                            |                     | 0.043***<br>(0.001)  | 0.043***<br>(0.001)  | 0.044***<br>(0.001)   | 0.043***<br>(0.001)  | 0.044***<br>(0.001)  |
| Missing rating                    |                     | 0.004<br>(0.004)     | 0.004<br>(0.004)     | 0.013**<br>(0.004)    | 0.004<br>(0.004)     | 0.013**<br>(0.004)   |
| Log reviews                       |                     | -0.095***<br>(0.002) | -0.095***<br>(0.002) | -0.085***<br>(0.002)  | -0.095***<br>(0.002) | -0.085***<br>(0.002) |
| Superhost                         |                     | 0.072***<br>(0.003)  | 0.072***<br>(0.003)  | 0.076***<br>(0.003)   | 0.072***<br>(0.003)  | 0.076***<br>(0.003)  |
| Instantly bookable                |                     | 0.076***<br>(0.002)  | 0.076***<br>(0.002)  | 0.067***<br>(0.002)   | 0.076***<br>(0.002)  | 0.067***<br>(0.002)  |
| Log minimum nights                |                     | -0.045***<br>(0.001) | -0.045***<br>(0.001) | -0.049***<br>(0.001)  | -0.045***<br>(0.001) | -0.049***<br>(0.001) |
| Availability 365                  |                     | 0.034***<br>(0.001)  | 0.034***<br>(0.001)  | 0.037***<br>(0.001)   | 0.034***<br>(0.001)  | 0.037***<br>(0.001)  |
| Airbnb density                    |                     |                      | -0.011<br>(0.017)    | -0.042<br>(0.022)     | -0.008<br>(0.017)    | -0.046*<br>(0.022)   |
| Share entire home                 |                     |                      | 0.044***<br>(0.007)  | 0.035***<br>(0.009)   | 0.043***<br>(0.007)  | 0.036***<br>(0.009)  |
| Mean neighborhood reviews         |                     |                      | 0.012<br>(0.011)     | 0.032*<br>(0.014)     | 0.012<br>(0.011)     | 0.032*<br>(0.014)    |
| Mean neighborhood availability    |                     |                      | 0.008<br>(0.010)     | 0.025<br>(0.013)      | 0.009<br>(0.010)     | 0.025<br>(0.013)     |
| Tourist-amenity density           |                     |                      | 0.110***<br>(0.015)  | 0.129***<br>(0.020)   | 0.110***<br>(0.015)  | 0.129***<br>(0.020)  |
| Distance to the center            |                     |                      | -0.038***<br>(0.012) | -0.043**<br>(0.016)   | -0.038***<br>(0.012) | -0.042**<br>(0.016)  |
| Entire home × Airbnb density      |                     |                      |                      |                       | -0.005<br>(0.003)    | 0.033**<br>(0.012)   |
| SD neighborhood intercept         | 0.254               | 0.204                | 0.123                | 0.157                 | 0.123                | 0.158                |
| SD neighborhood entire-home slope |                     |                      |                      | 0.210                 |                      | 0.207                |
| SD city intercept                 | 0.248               | 0.253                | 0.270                | 0.284                 | 0.270                | 0.284                |
| SD residual                       | 0.583               | 0.427                | 0.427                | 0.414                 | 0.427                | 0.414                |
| Observations                      | 143159              | 143159               | 143159               | 143159                | 143159               | 143159               |
| Marginal R <sup>2</sup>           | 0.000               | 0.363                | 0.424                | 0.410                 | 0.424                | 0.426                |
| Conditional R <sup>2</sup>        | 0.271               | 0.597                | 0.612                | 0.645                 | 0.612                | 0.653                |
| AIC                               | 252775.3            | 163633.2             | 163411.6             | 155585.1              | 163420.4             | 155587.0             |
| BIC                               | 252814.8            | 163781.3             | 163618.9             | 155802.3              | 163637.6             | 155814.0             |
| ICC                               | 0.3                 | 0.4                  | 0.3                  | 0.4                   | 0.3                  | 0.4                  |
| RMSE                              | 0.58                | 0.43                 | 0.43                 | 0.41                  | 0.43                 | 0.41                 |

Note: the dependent variable is logged nightly price. Standard errors in the parentheses. Continuous predictors are standardized. M0 is the empty model; M1 adds listing-level controls; M2 adds neighborhood context including tourist-amenity density; M3 adds a neighborhood-level random slope for entire-home status; M4 adds the entire-home by Airbnb density interaction; M5 combines the random slope and interaction. \* $p < 0.05$ ; \*\* $p < 0.01$ ; \*\*\* $p < 0.001$ .